

Infants learn saltatory alternations: evidence from Catalan

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Catalan Phonology

Catalan has final devoicing:

$[-\text{sonorant}] \rightarrow [-\text{voice}] / ______]_{\text{word}}$

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Example:

Note that the masculine is marked with no ending and feminine with $[-\text{ə}]$

‘grey.M.SG’	‘grey-F.SG’	‘fat.M.SG’	‘fat-F.SG’	
/griz/	/griz-ə/	/gras/	/gras-ə/	underlying representations
gris	—	—	—	Final Devoicing
[^l gris]	[^l grizə]	[^l gras]	[^l grasə]	surface representations

/ʒ/-final stems

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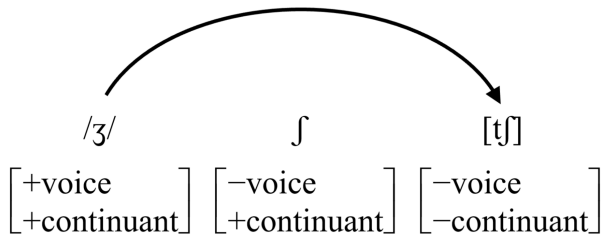
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Rule-based phonology: [ʒ] → [−continuant] / ____]_{word}

<i>'crazy.F.SG'</i>	<i>'crazy.M.SG'</i>	<i>'grey.M.SG'</i>	<i>'loose.M.SG'</i>	
/bɔʒ-ə/	/bɔʒ/	/griz/	/fluʃ/	underlying representations
—	bɔdʒ	—	—	/ʒ/-affrication
—	bɔtʃ	gris	—	Final Devoicing
['bɔʒə]	['bɔtʃ]	['gris]	['fluʃ]	surface representations

Saltation

This type of alternation is referred to as **saltation** (from Latin *saltus* 'leap') (Hayes & White, 2015)



The challenge of saltation

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Theory

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Learnability

Experimental evidence that saltation is unnatural and liable to repair

Saltation and OT

Łubowicz (2002) and Ito and Mester (2003) show that saltation **cannot be derived** in classical OT

‘classical’ OT in the sense of Prince and Smolensky (1993) with the faithfulness constraints of McCarthy and Prince (1995)

Demonstrating the impossibility of saltation: deriving [bɔtʃ] from /bɔʒ/

Markedness constraint for final devoicing (*[-son, +voice]#) must outrank faithfulness constraint for voicing (IDENT[voice])

Demonstrating the impossibility of saltation: deriving [bətʃ] from /bətʃ/

Markedness constraint for final devoicing (*[-son, +voice]#) must outrank faithfulness constraint for voicing (IDENT[voice])

Markedness constraint against final palatal fricatives (FORT(pal)#) must outrank faithfulness constraint for continuancy (IDENT[cont])

FORT(pal)# assesses a violation for every word-final palatal fricative (e.g. ʃ# or ʒ#)

Saltation leads to a ranking contradiction

/bɔʒ/	*[-son,+voi]#	FORT(pal)#	ID[voi]	ID[cont]
☞ a. bɔtʃ			*	*
b. bɔʃ		*!	*	
c. bɔʒ	*!	*		

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/bɔʒ/	*[-son,+voi]#	FORT(pal)#	ID[voi]	ID[cont]
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b. bɔf		*!	*	
c. bɔʒ	*!	*		

/fluʃ/	*[-son,+voi]#	FORT(pal)#	ID[voi]	ID[cont]
☞ a. flutʃ				*
✗ b. fluʃ		*!		

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adults (White, 2014)

and infants (White & Sundara, 2014)

Saltation and Catalan

Previous evidence

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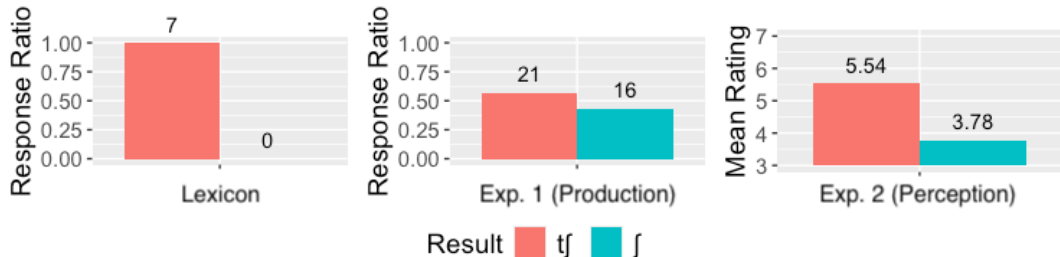


Figure: Results from Liang et al. (under review)

Saltation in Infancy

Can infants with long term exposure learn a saltatory alternation?

If saltation is so difficult to learn, we should see evidence of this in infants learning a language with saltation as well

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No study to date has investigated whether infants learning a language with a saltatory alternation can learn it productively

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Catalan Infant-Directed Speech Input

Catalan infants hear few word types exhibiting saltation, but they are high-frequency and likely to be a part of their early lexicons

Across all 7 CHILDES Catalan corpora (146k words):

- 7 word types with saltation

- 315 [tʃ] tokens and 64 [ʒ] tokens, representing 0.2% of the corpora

Experiment 1: do infants learn saltation?

Research Question

Do Catalan-learning infants extend saltatory alternations to novel forms?

Experiment 1 Participants

20 Catalan-dominant 24-month-olds (mean = 724 days, S.D. = 161 days)

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Recruited through various preschools in the Barcelona area

Procedure

Central Fixation Procedure (Cohen, 1972) adapted to school setting

Habit software (Oakes, Sperka, DeBolt, & Cantrell, 2019)



Figure: Sample experimental setup

Experimental Paradigm

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All recordings were done by a native speaker in a soundproof booth

Experiment 1: Two sub-groups

[kɛʒə], [ʌʒə] **condition**

Familiarization:

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If they extend saltation to novel forms:

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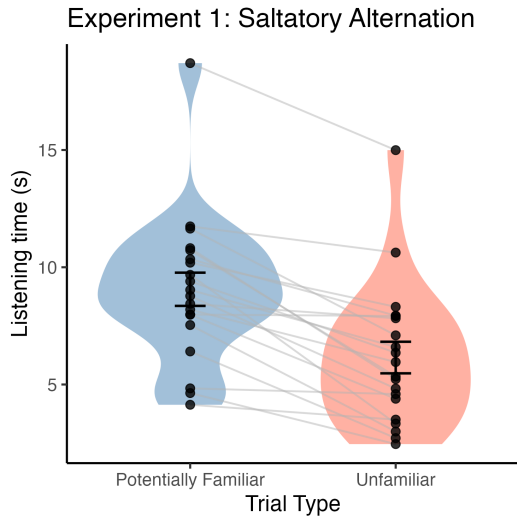
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From Experiment 1, Catalan-learning infants actually do learn saltation and generalize it to novel forms

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Research Question

Do Catalan-learning infants also “repair” saltation by simplifying the pattern to straightforward final devoicing as adults do?

Experiment 2 Participants and Procedure

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Identical screening, recruitment, and experimental procedures to Experiment 1

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Results (Experiments 1 + 2)

Model

$LT \sim \text{Block} \times \text{Experiment} \times$
 $\text{Trial Type} + (1 + \text{Block} +$
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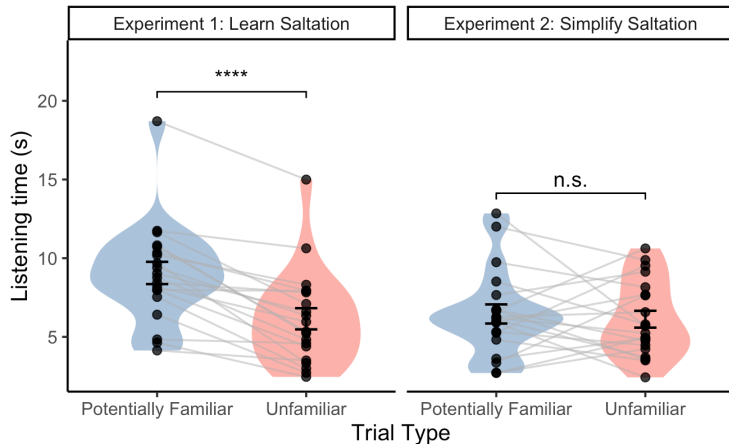
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Key Interaction

Experiment $\times$ Trial Type:
 $F(1, 183) = 9.68,$
 $p = .002$



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Conclusion

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Infant Catalan-acquirers seemingly **outperform even adult Catalan speakers** in generalizing saltatory alternations

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As the lexicon grows, the percentage of the lexicon that consists of saltatory forms decreases along with willingness to generalize



Moltes gràcies!
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But [dɜ] never surfaces → still hard to learn (McCarthy, 2005)